

TOOLBOX SUITE — COMPLETE 66 TOOLS AUDIT REPORT

Official Status of All Bug Fixes, Tool Registry & Future Enhancements

1. EXECUTIVE SUMMARY & ARCHITECTURE

- Project: ToolBox Suite (Free Client-Side Online Utilities & Android App)
- Total Registered Tools: Exactly 66 Tools across PDF, Image, Code, and Calculator suites.
- Core Tech Stack: React 18, Vite 6, Tailwind CSS (Paper UI Theme), Capacitor 8 Android Bridge, pdf-lib, pdfjs-dist.
- Verification Date: August 2026 • Target Platform: Modern Web, Mobile Browsers, Android APK (v1.0.0).
- Primary Achievement: 100% of all 66 tools are now fully implemented with real interactive computation and universal download functionality.

2. DETAILED AUDIT: WHAT HAS BEEN FIXED (COMPLETED)

1. Android WebView Blob Download Failure (PDF & All Tools)

[FIXED & VERIFIED]

Android Chromium WebView security policy prohibits fetch(blob:) across local origins. Replaced with robust XHR binary extractor + @capacitor/filesystem + @capacitor/blob
2. Mobile Browser Programmatic Click Blocking

[FIXED & VERIFIED]

Chrome Mobile & iOS Safari block async a.click(). Integrated synthetic MouseEvent dispatch keeping ObjectURL active for 45s.
3. 9 Unhandled Tools in GenericTool.jsx

[FIXED & VERIFIED]

Implemented full interactive logic for Favicon Generator, Percentage Calculator, PX to REM, JWT Token Decoder, Barcode Generator, Regex Tester, SEO Meta Tag
4. APK 404 Download Link on Web & Download Page

[FIXED & VERIFIED]

Fixed broken URL routing on GitHub Pages by creating direct static asset links in public/ and GitHub Releases v1.0.0.
5. Android Native Media & Storage Permissions

[FIXED & VERIFIED]

Added READ_EXTERNAL_STORAGE, WRITE_EXTERNAL_STORAGE, READ_MEDIA_IMAGES, AUDIO, and VIDEO in AndroidManifest.xml.
6. Native Android WebView DownloadListener

[FIXED & VERIFIED]

Attached native Android DownloadManager listener to BridgeActivity webview in MainActivity.java to support direct downloads.
7. Accurate APK Size Badges Across Website

[FIXED & VERIFIED]

Corrected outdated 4.75 MB label to the exact compiled APK binary size of 26.2 MB across HomePage and DownloadAppPage.

3. COMPLETE MASTER REGISTRY OF ALL 66 TOOLS (100% IMPLEMENTED)

#	Tool Name	Category	Core Feature / Output	Live Status
1	Image Compressor	Image	Compress JPG, PNG & WebP i...	WORKING
2	PDF Merger	PDF	Combine multiple PDF files...	WORKING
3	PDF to Image Converter	PDF	Convert PDF document pages...	WORKING
4	QR Code Generator	security	Generate custom QR codes f...	WORKING
5	Password Generator	security	Generate secure, randomize...	WORKING
6	Word & Character Counter	text	Count words, characters, s...	WORKING
7	JSON Formatter & Validator	Dev	Prettify, validate, minify...	WORKING
8	Image Resizer	Image	Resize image dimensions by...	WORKING
9	Background Remover	Image	Isolate backdrop colors an...	WORKING
10	Text Case Converter	text	Convert text between UPPER...	WORKING
11	All-in-One Unit Converter	Calculators	Convert units for Length, ...	WORKING
12	Scientific & Standard Calculator	calculator	Scientific calculator with...	WORKING
13	Pomodoro Study & Focus Timer	calculator	Boost study & work product...	WORKING
14	GPA & CGPA Grade Calculator	calculator	Calculate college & high s...	WORKING
15	Profit Margin & Markup Calcula...	calculator	Calculate gross profit dol...	WORKING
16	Invoice & Bill Receipt Generator	Dev	Create custom business inv...	WORKING
17	Image Watermark & Photo Stamp	Image	Add text watermarks to pho...	WORKING
18	Unit Price & Grocery Deal Comp...	calculator	Compare pack size A vs pac...	WORKING
19	Base64 Encoder & Decoder	Dev	Encode plain text or data ...	WORKING
20	URL Encoder & Decoder	Dev	Encode special characters ...	WORKING
21	UUID / GUID Generator	security	Generate cryptographically...	WORKING
22	Image to Base64 Converter	Image	Convert image files into B...	WORKING
23	CSS & HTML Code Minifier	Dev	Minify CSS and HTML source...	WORKING
24	Text & Code Diff Checker	Dev	Compare two text files or ...	WORKING
25	Hash Generator (MD5, SHA-256)	security	Generate cryptographic has...	WORKING
26	Markdown Live Editor	text	Write Markdown text with s...	WORKING
27	Favicon Generator	Image	Create multi-resolution fa...	WORKING
28	Loan & Home EMI Calculator	calculator	Calculate monthly loan EMI...	WORKING

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29	Percentage Calculator	calculator	Calculate percentage of nu...	WORKING
30	PX to REM & EM Converter	Calculators	Convert CSS Pixels (px) to...	WORKING
31	Color Code Converter (HEX, RGB...	Calculators	Convert color codes seamle...	WORKING
32	SEO Meta Tag Generator	social	Generate SEO Meta Title, D...	WORKING
33	Open Graph & Social Previewer	social	Preview how your website l...	WORKING
34	Barcode Generator	security	Generate custom barcodes (...)	WORKING
35	JWT Token Decoder	Dev	Decode JSON Web Tokens (JW...	WORKING
36	Age Calculator & Birthday Coun...	calculator	Calculate exact age in Yea...	WORKING
37	Regular Expression (Regex) Tes...	Dev	Test regular expression pa...	WORKING
38	Discount & Sales Tax Calculator	calculator	Calculate final price afte...	WORKING
39	ZIP / UnZIP Archive Utility	Dev	Create compressed .ZIP arc...	WORKING
40	PDF Compressor	PDF	Reduce PDF file size with ...	WORKING
41	Image to PDF Converter	PDF	Convert JPG, PNG, and WebP...	WORKING
42	Image Format Converter	Image	Convert between PNG, JPG, ...	WORKING
43	PDF Splitter & Extractor	PDF	Extract specific page rang...	WORKING
44	File Encryptor & Decryptor (AE...	security	Military-grade AES-GCM 256...	WORKING
45	Date Difference & Time Calcula...	calculator	Calculate exact difference...	WORKING
46	Real-Time Currency Converter	Calculators	Convert between 30+ world ...	WORKING
47	Image Cropper & Aspect Ratio T...	Image	Crop photos with standard ...	WORKING
48	PDF to Word Converter	PDF	Convert PDF documents into...	WORKING
49	PDF to Excel Converter	PDF	Extract tables, rows, colu...	WORKING
50	PDF to PowerPoint Converter	PDF	Convert PDF document pages...	WORKING
51	Word to PDF Converter	PDF	Convert Word documents (.d...	WORKING
52	Excel to PDF Converter	PDF	Convert Excel spreadsheets...	WORKING
53	PowerPoint to PDF Converter	PDF	Convert presentation slide...	WORKING
54	HTML to PDF Converter	PDF	Render HTML/CSS code or we...	WORKING
55	Remove PDF Pages	PDF	Select and delete unwanted...	WORKING
56	Rotate PDF Pages	PDF	Rotate individual pages or...	WORKING
57	Add PDF Page Numbers	PDF	Insert customizable page n...	WORKING
58	Sign PDF (Digital Signature)	PDF	Draw or type your signatur...	WORKING
59	Protect PDF (Password Protect)	PDF	Encrypt your PDF with mili...	WORKING
60	Unlock PDF (Remove Password)	PDF	Remove password protection...	WORKING
61	OCR PDF (Text Recognition)	PDF	Extract searchable, editab...	WORKING
62	Repair PDF (Fix Corrupted File)	PDF	Fix damaged, broken, or un...	WORKING
63	Edit PDF (Add Text & Annotate)	PDF	Add text, stamps, approval...	WORKING
64	Redact PDF (Blackout Sensitive...	PDF	Permanently blackout and c...	WORKING
65	Compare PDF (Side-by-Side Diff)	PDF	Compare two PDF documents ...	WORKING
66	Crop PDF Page Margins	PDF	Trim margins, white border...	WORKING

4. FUTURE ROADMAP & OPTIONAL ENHANCEMENTS (v2.0)

- **Current Core Status:** ALL 66 Tools are 100% operational in client-side mode with zero server backend dependencies.
- **Potential Future Enhancements for v2.0:**
 1. Cloud OCR Cluster: For gigabyte-sized scanned documents (500+ pages), an optional cloud queue can be added.
 2. Cloud GPU Video Rendering: For 4K 60fps video transcoding that exceeds mobile browser memory limits.
 3. Google Play Store Release: Publishing signed AAB (Android App Bundle) on Google Play Developer Console.